PHY202: Problem Set 4

NOTE: Submit the solution to problem number 1 in the beginning of the class on Oct. 6th.

1. Find the Fourier transform of the following function,

$$f(x) = \begin{cases} 2a - |x|, & |x| < 2a \\ 0, & |x| > 2a \end{cases}$$

and use Parseval's theorem to evaluate $\int_0^\infty \frac{\sin^4(ak)}{k^4}\, dk$.

2. Find the Fourier transform of the following functions:

(i) $f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise} \end{cases}$

(ii) $f(x) = \begin{cases} x, & 0 < x < 1, \\ 0, & \text{otherwise} \end{cases}$

(iii) $f(x) = \begin{cases} \cos x, & |x| < \pi/2, \\ 0, & \text{otherwise} \end{cases}$

3. Consider the differential equation given below ($x' \equiv dx/dt$, etc.):

$$x''(t) + ax'(t) + bx(t) = f(t).$$

Outline the general approach to solve this using Fourier transform. Make any specific choice for a, b and $f(t)$ to illustrate your general procedure for solving such differential equations.